EE320 Lab2 — State Space Representation

University of Strathclyde Engineering

- In Matlab (Octave, or other numerical tools), you can enter and algebraically manipulate matrices;
- to enter the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

you can type

```
A = [1 2 3; 4 5 6];
```

or

```
A = [1 2 3; <enter>
   4 5 6];
```

- when multiplying matrices $\mathbf{AB}$, you need to ensure that dimensions fit: the number rows in $\mathbf{B}$ has to be the same as the number of columns in $\mathbf{A}$;
- to calculate the eigenvalue decomposition of a square matrix, check out the command `eig()`, e.g. by typing `help eig`;
- for inverting a square matrix, please see the command `inv()`.

Exercise 1

- You are given the following system

$$\begin{bmatrix} \dot{z}_1(t) \\ \dot{z}_2(t) \\ \dot{z}_3(t) \\ \dot{z}_4(t) \end{bmatrix} = \begin{bmatrix} 5 & 1 & -2 & 0 \\ 1 & 5 & 0 & -2 \\ -2 & 0 & 5 & 1 \\ 0 & -2 & 1 & 5 \end{bmatrix} . \begin{bmatrix} z_1(t) \\ z_2(t) \\ z_3(t) \\ z_4(t) \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 2 & -1 & -3 \\ 0 & -3 & -1 \\ 0 & -1 & 3 \end{bmatrix} . \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} z_1(t) \\ z_2(t) \\ z_3(t) \\ z_4(t) \end{bmatrix} \quad (1)$$

- check the controllability and observability of this system:
  1. can every input control every state?
  2. are there states that cannot be controlled from any input (i.e. is the system not controllable)?
  3. can every state be observed from every output?
  4. are there states that cannot be observed from any output (i.e. is the system not observable)?

Exercise 2

- You are given the following system

$$\begin{bmatrix} \dot{z}_1(t) \\ \dot{z}_2(t) \\ \dot{z}_3(t) \end{bmatrix} = \begin{bmatrix} 2 & \frac{2}{3} & \frac{1}{3} \\ 1 & \frac{5}{3} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{7}{3} \end{bmatrix} \cdot \begin{bmatrix} z_1(t) \\ z_2(t) \\ z_3(t) \end{bmatrix} + \begin{bmatrix} \frac{2}{3} & 0 \\ -\frac{1}{3} & -1 \\ \frac{2}{3} & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$\begin{bmatrix} y_1(t) \end{bmatrix} = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} z_1(t) \\ z_2(t) \\ z_3(t) \end{bmatrix} \qquad (2)$$

- check the controllability and observability of this system:
  1. can every input control every state?
  2. can every state be observed from every output?
  3. are there states that cannot be observed from any output?

Exercise 3

- You are given the following system

$$\begin{bmatrix} \dot{\hat{z}}_1(t) \\ \dot{\hat{z}}_2(t) \\ \dot{\hat{z}}_3(t) \end{bmatrix} = \begin{bmatrix} \frac{5}{2} & -\frac{1}{2} & -\frac{1}{6} \\ -1 & 2 & -\frac{1}{3} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{3}{2} \end{bmatrix} \cdot \begin{bmatrix} \hat{z}_1(t) \\ \hat{z}_2(t) \\ \hat{z}_3(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{6} & -\frac{1}{2} \\ -\frac{2}{3} & 0 \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$\begin{bmatrix} y_1(t) \end{bmatrix} = \begin{bmatrix} 3 & -3 & -1 \end{bmatrix} \cdot \begin{bmatrix} \hat{z}_1(t) \\ \hat{z}_2(t) \\ \hat{z}_3(t) \end{bmatrix} \tag{3}$$

- Can you demonstrate that this system is identical to the one in exercise 2?
- Hint: try to bring both systems into the same defined form (direct form I, II, etc)!